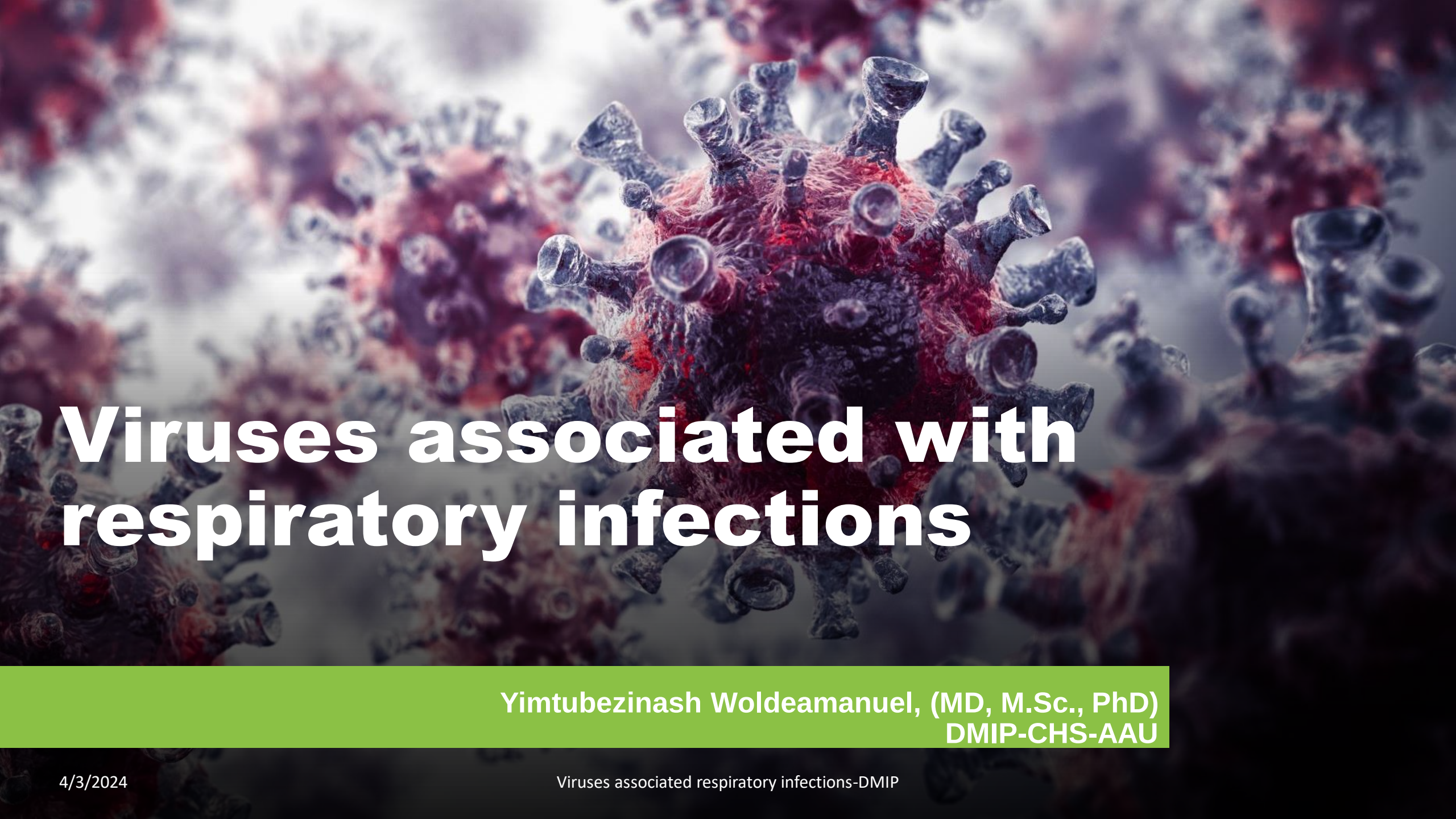




Viruses associated with respiratory infections

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Outline of Lecture

Introduction

Influenza virus

Parainfluenza virus

Measles virus

Corona virus

Others...

Introduction

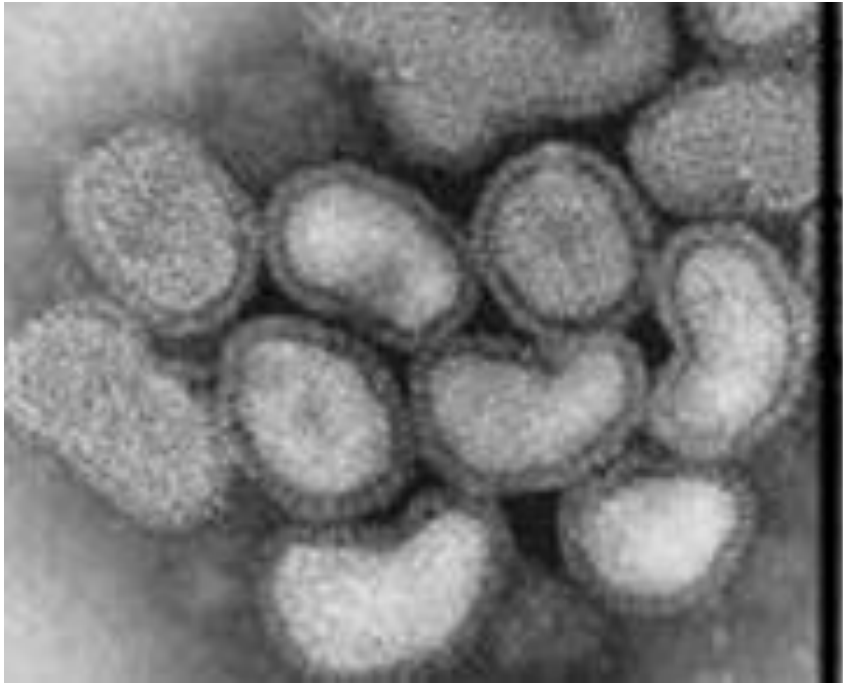
- Most upper respiratory infections , including middle and lower respiratory infections are caused by viruses
 - Most common :- influenza viruses, parainfluenza viruses, rhinoviruses, adenoviruses, respiratory syncytial virus (RSV), and coronaviruses
- These groups of heterogenous viruses share some characteristics, such as short incubation periods and person to person transmission
- Transmission is direct, by infective droplet nuclei, or indirect by hand transfer of contaminated secretions to nasal or conjunctival epithelium

Influenza virus

Family : **Orthomyxoviridae**
Type: ss RNA, linear, segmented, negative sense
Shape: usually spherical; pleomorphic; enveloped
Size: 80-120 nm
Nucleocapsid: helical

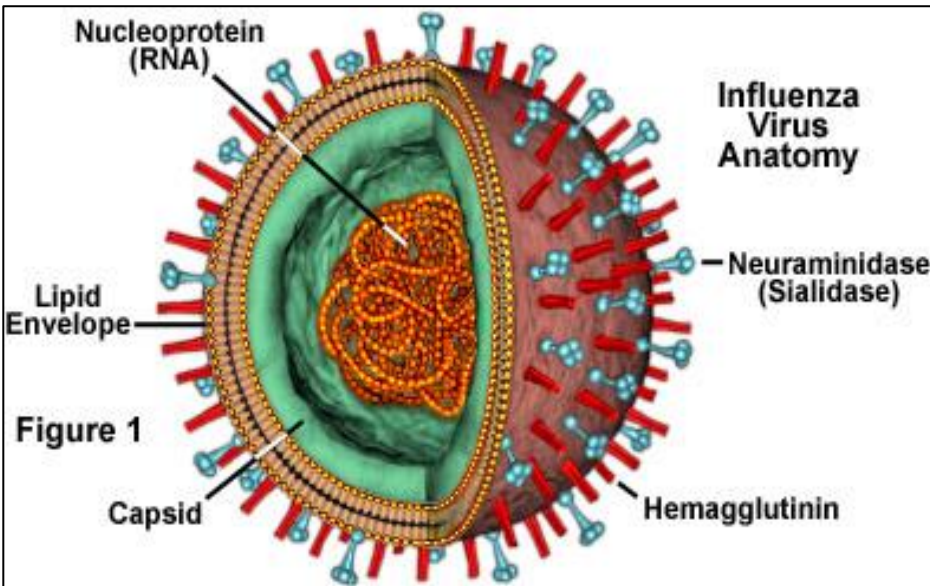
Three immunological types: Influenza A, B & C

- Influenza A infects humans and animals
- Influenza B infects only humans
- Influenza C is not a major human pathogen



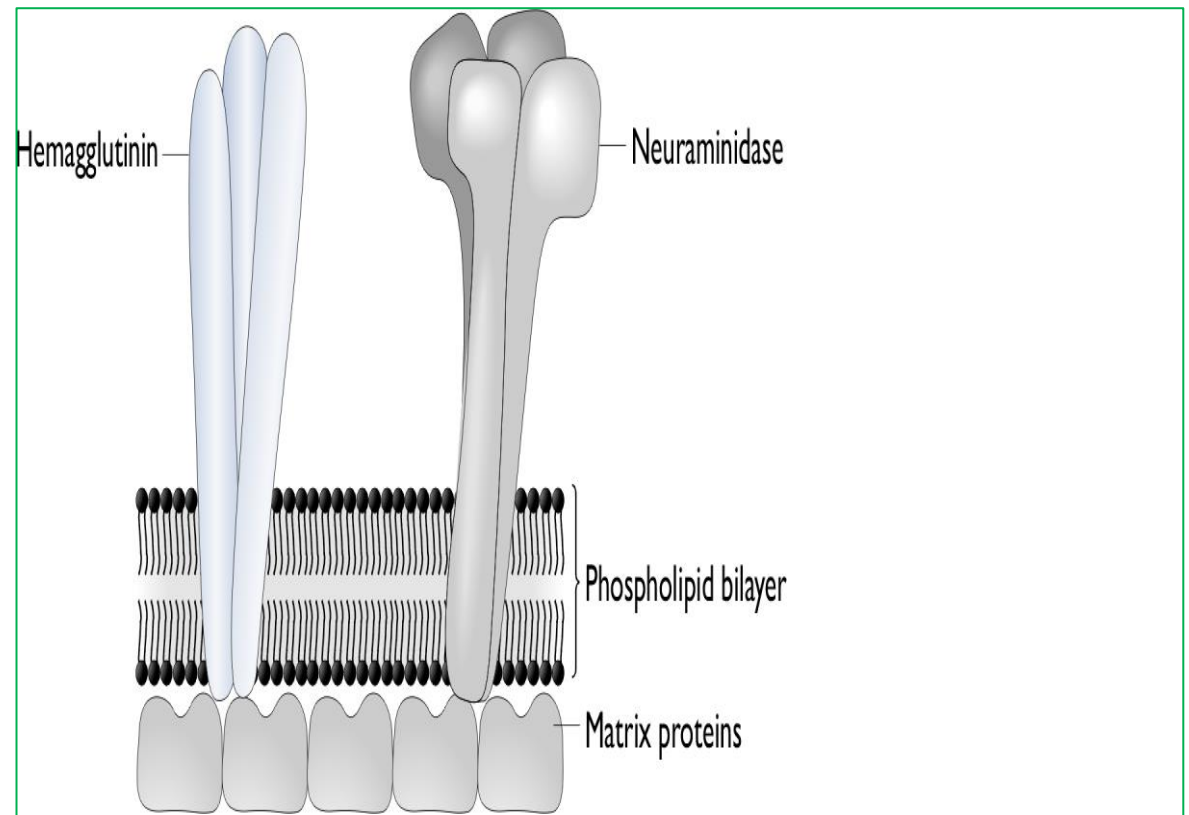
Structure

- **Virion:** usually spherical, enveloped
- **Genome:** Influenza A & B have 8 segments; C has 7
- **Proteins :**
 - 7 structural proteins (PB1, PB2, PA, NP, M1, HA, M2)
 - 3 nonstructural (NS1, NS2, M2) observed in infected cells
- **Envelope:-**
 - glycolipid membrane, two virus specific glycoproteins : hemagglutinin (HA) and neuraminidase (NA); Influenza C has no NA



Structure and function of Hemagglutinin (HA)

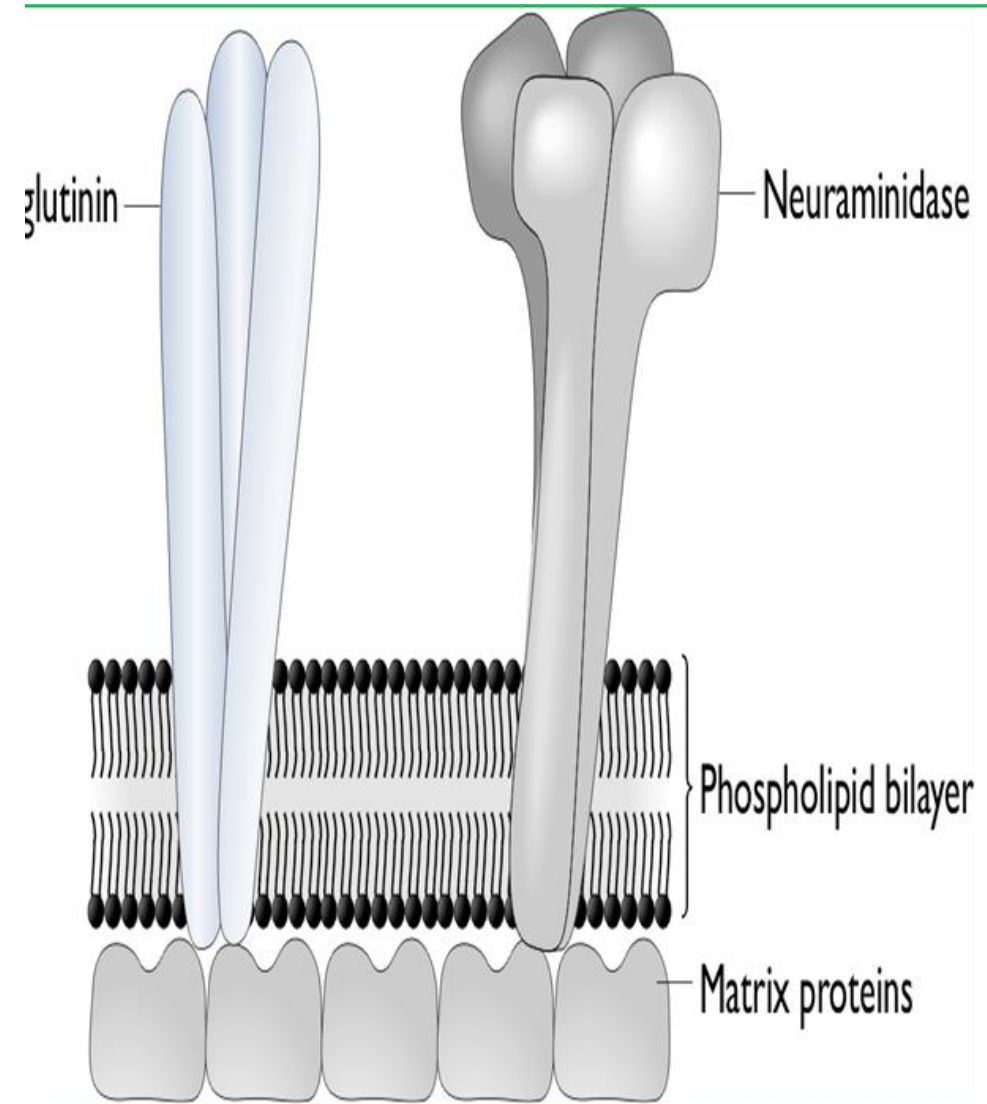
- 13 subtypes (H1-H13); 3 in humans
- five antigenic sites on the HA molecule
- HA spike on the viral particle is a trimer (3 intertwined HA1 & HA2 dimers)



- Extracellular proteases – cleave HA
- Fusion by amino terminus of HA2
- ability to agglutinate RBC from certain species in vitro
- serves as a point of attachment to N-acetylneuraminic acid-containing glycoprotein or glycolipid receptor sites on human respiratory cell surfaces

Structure and function of Neuraminidase (NA)

- Slender stalk topped with a box shaped head
- catalytic site for NA on top of each head
- removes sialic acid residues from viral glycoprotein
- nine subtypes (N1-N9; two in humans)



Antigenic variations

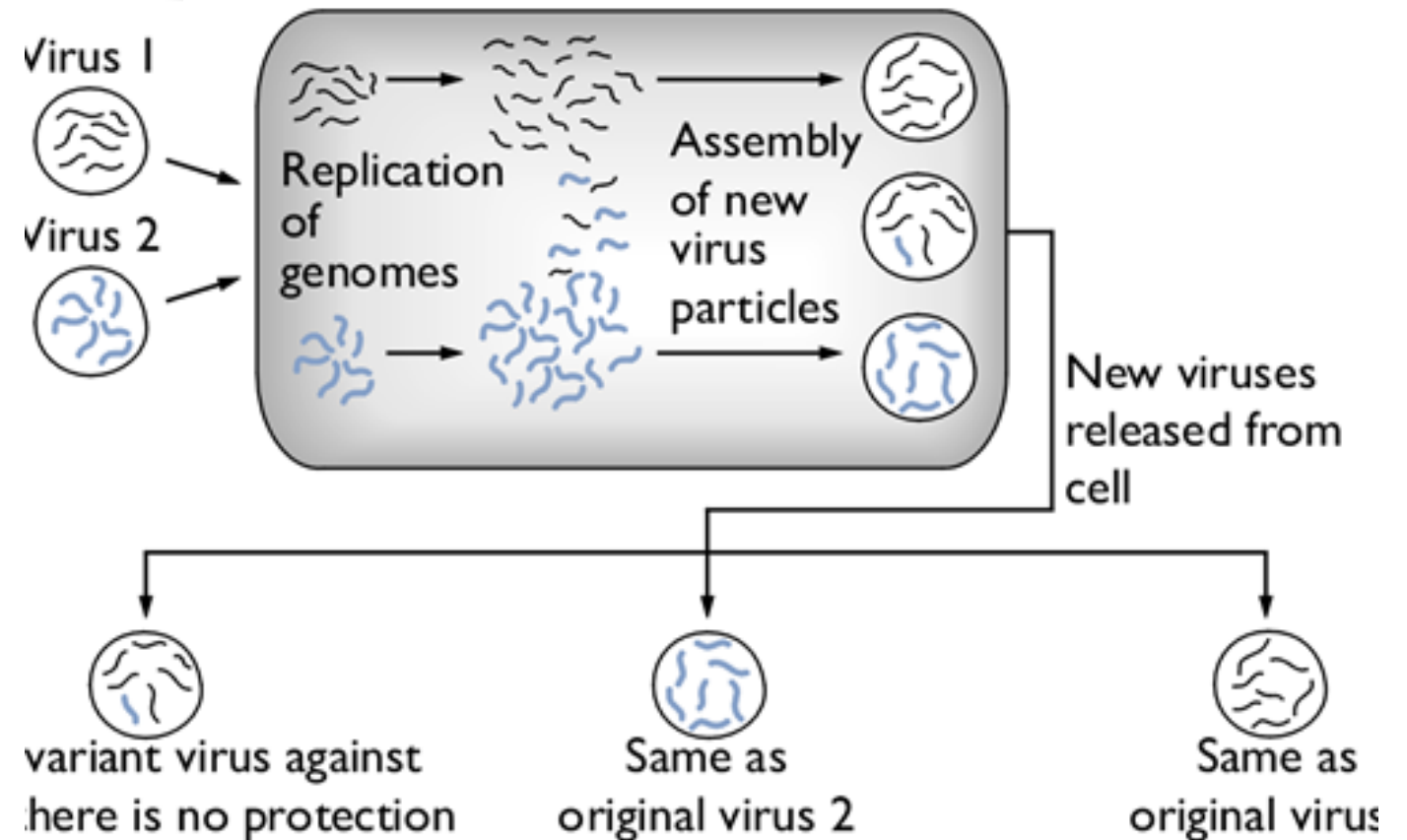
1. Antigenic drift

- Small change in viral antigens
- Occurs in all types (A, B and C)
- Due to RNA Polymerases (Replicases) which are inherently inaccurate
 - Results in random mutations and rapid evolution
- Mutant hemagglutinin generated may escape full antibody protection. Those viruses expressing mutant hemagglutinin expand and spread to other people
- Causes epidemics

2. Antigenic Shift

- Major change in viral antigens
- Occurs in “A” type only (“A” type infects humans and animals)
- Due to reassortment and mixing of RNA segments
 - Virus acquires new Hemagglutinin (H) or Neuraminidase (N) and population does not have antibodies against that new antigen for protection; causes pandemics

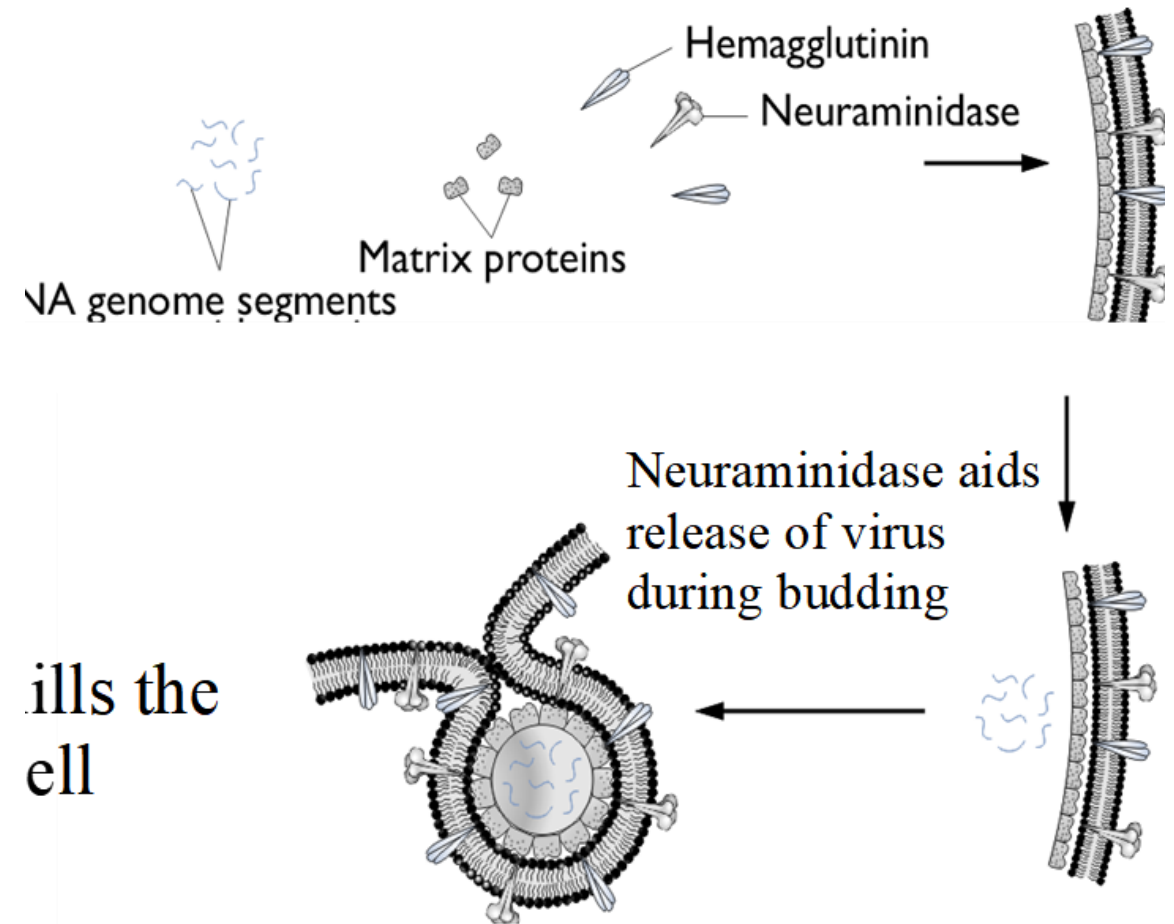
Antigenic Shift by reassortment (Influenza A)



Replication

Hemagglutinin aids binding of virus to host cell, surface sialic acid

- penetration by receptor mediated endocytosis
- fusion between virus envelope and membrane of phagolysosome
- Uncoating of nucleoprotein into host cytoplasm
- Viral polymerase transcribes genome into 8 mRNAs, and is translated
- Progeny RNA synthesized in the nucleus (by viral encoded polymerase: P1, P2, P3) and nucleocapsid assembled in the cytoplasm
- In the cell wall, viral matrix protein + nucleocapsid + viral envelope; maturation by budding



Epidemiology

- Direct droplet spread & direct contact
- Incubation 1 to 4 days
- Major antigenic shift correlates with epidemics and minor antigenic drifts with maintenance in the population
- Humans are major hosts

Pathogenesis

- Virus multiplies in respiratory epithelium
- Cilia damage and cell desquamation
- Mechanical clearance of respiratory tract compromised
- Inflammation is caused by viral toxicity
- Phagocytic host defense impaired
- Susceptibility to secondary bacterial invaders



Influenza victims crowd into an emergency hospital near Fort Riley, Kansas in 1918. # National Museum of Health / AP

Source:

<https://www.theatlantic.com/photo/2018/04/photos-the-1918-flu-pandemic/557663/>

Pandemics :-

- 1918 “Spanish Flu” >20 million dead worldwide
 - “Hidar Beshita” in Ethiopia
- 1957 “Asian Flu”
- 1968 “Hong Kong Flu”

Disease

- Influenza A & B tend to cause the most severe illness
- Acute disease with dry cough , fever, (“Flu–like” Symptoms)
- cough & weakness persist for 1-2 weeks

Complications of Influenza

- Post–influenza pneumonia by bacteria
 - *Haemophilus influenzae*; *Streptococcus pneumoniae* etc.
- Viral or combination
- Reye’s syndrome , acute encephalopathy (2–16-year-olds); Fatty infiltration of liver & brain edema; possible exposure to salicylates; high mortality rate
- Cardiovascular complications :- myocarditis, acute myocardial infarction, and exacerbation of heart failure (HF)

Laboratory diagnosis

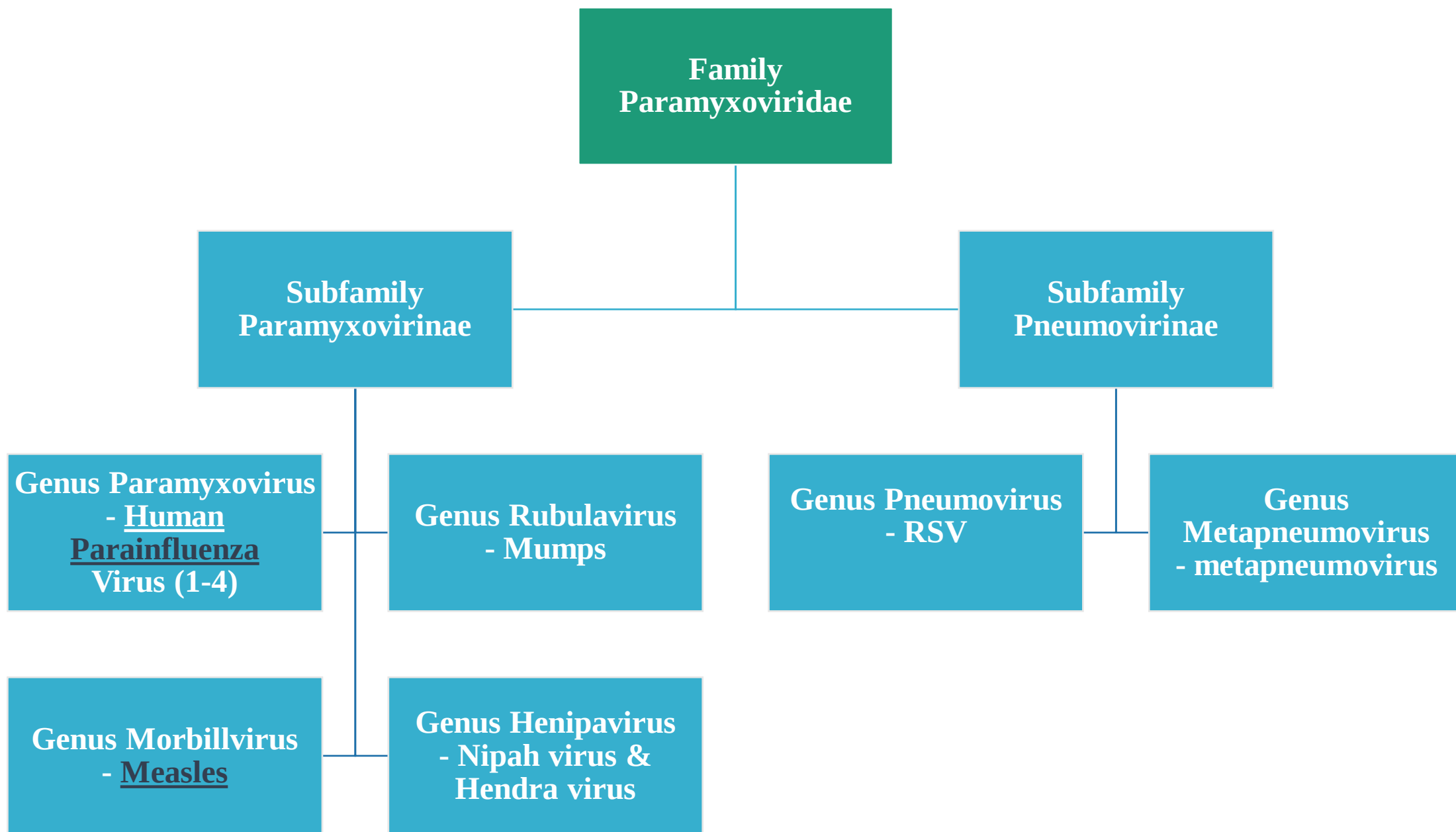
- Specimen: nasal washing & throat swaps
- Isolation: Primary monkey kidney cell culture
 - Amniotic cavity of embryonated eggs
 - Continuous cell lines derived from canine kidney
 - (MDCK) or rhesus monkey kidney (LLC-MK₂)
- Incubation: addition of trypsin; without serum
- Examination: Hemagglutination of culture fluid on the 7th day
- Identification: hemagglutination inhibition; Hemadsorption , Direct immunofluorescence detection
- Serology : Complement fixation; hemagglutination inhibition

Prevention

- High risk groups, must be vaccinated every year to keep up with antigenic drift/shift
- Governments agencies & WHO (World Health Organization) monitor currents strains
- Vaccine: Usually a cocktail containing type A and B
- Killed influenza virus injected intramuscular

Immunity to Influenza

- Antibodies against hemagglutinin are protective
- Antibodies against neuraminidase are helpful but not critical
- Immunity is long lived and subtype specific



Parainfluenza virus

<u>Family</u> :	Paramyxoviridae;
<u>Subfamily</u> :	Paramyxovirinae ; genus:-Paramyxovirus
<u>Type</u> :	ssRNA, non segmented, negative sense
<u>Envelope</u> :	+, from host cytoplasmic membrane
<u>Shape</u> :	pleomorphic
<u>Size</u> :	120-150 nm
<u>Nucleocapsid</u> :	helical
<u>Proteins</u> :	six structural proteins; 3 proteins participate in the formation of envelope
<u>Replication</u> :	in the cytoplasm, except for measles virus

Characteristic cytopathic effect (CPE):- formation of syncytia (multinucleated giant cells) & eosinophilic cytoplasmic inclusion bodies.

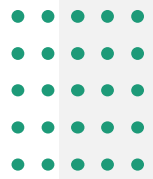
Parainfluenza virus infection

Common respiratory infection in persons of all ages

- Severe respiratory tract infection in infants and children
- Four serotypes (1-4)

Epidemiology:

- Transmission – person to person; large droplet aerosols
- Wide geographic distribution
- Type 3 most prevalent
- Reinfection common
- High risk places – pediatric wards; day care centers and schools



Pathogenesis

- Viremia uncommon
- Ranges from common cold to more extensive involvement of larynx and upper trachea croup (laryngotracheobronchitis type 1 & 2)
- pneumonia or bronchiolitis (type 3)
- primary infection – most severe
- antibodies from previous infection – modify course of illness
- incubation in adults 2-6 days

Clinical features

- Primary infection in young children- rhinitis, pharyngitis, fever, bronchitis
- Type 1 & 2 cause croup
- Type 3- bronchiolitis or pneumonia (infants less than 6 months)

Immunity

- Maternal antibodies
- IgA antibodies (disappear after a few months)
- Shared antigenic determinants among parainfluenza virus and mumps virus
- Serum antibodies to F protein more important



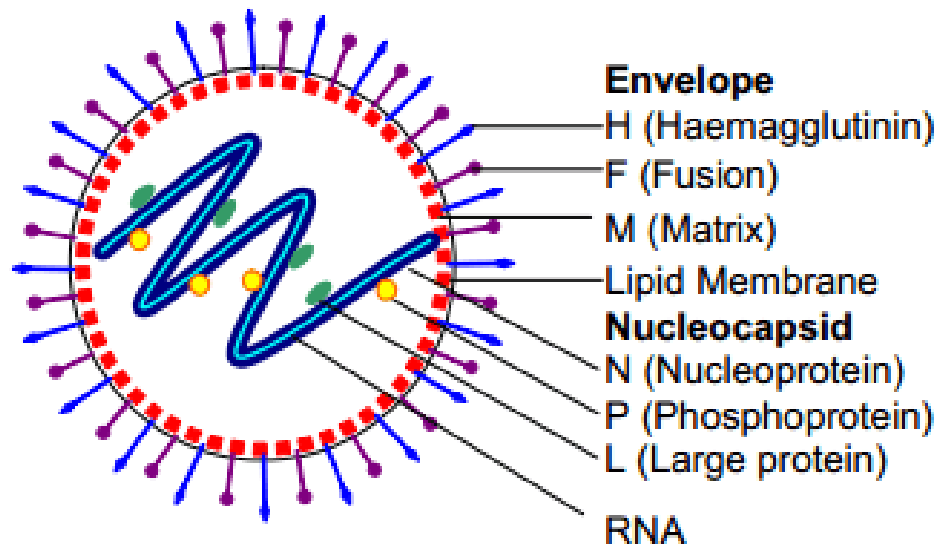
Laboratory diagnosis

- **Specimen** :- throat and nasal swabs
 - Prompt inoculation
- **Isolation**: primary human; continuous Monkey kidney cells lines (LLK-Mk₂)
 - Slow growth with minimum cytopathic effect
- **Detection**: hemadsorption
 - Typing: immunofluorescence or HI
 - Direct identification: IF or ELISA
- **Serology**- paired sera; Nt, ELISA or CFT (four-fold rise)

Measles virus

Family : Paramyxoviridae;
Subfamily: Paramyxovirinae
genus:- **Morbillivirus**

Transmission: person to person via respiratory secretions



Surface glycoprotein:

H binds to receptor and has hemagglutination activity; specific receptor for measles virus SLAM (signaling lymphocyte activation molecule) ; **F** mediates fusion (No neuraminidase activity)

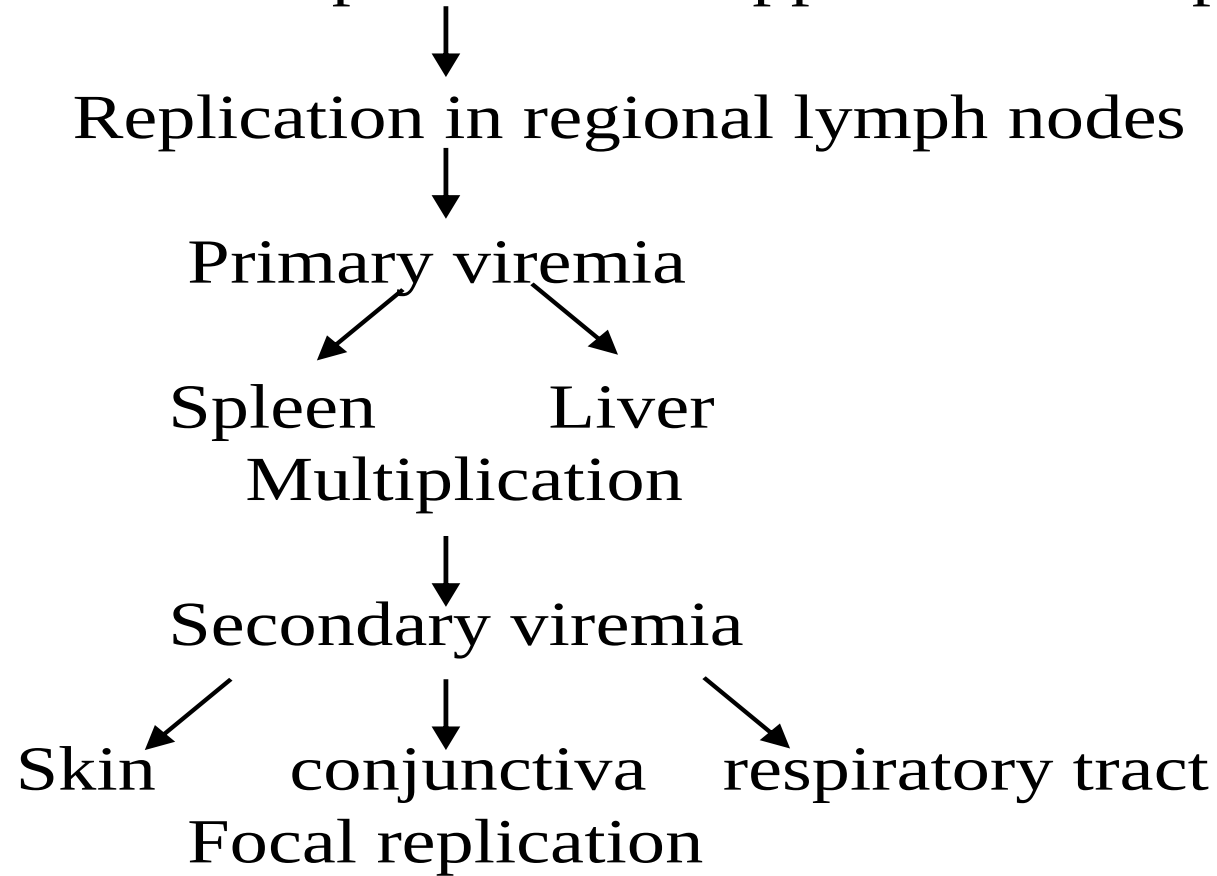
Measles (Rubeola) virus infection

Epidemiology

- Highly contagious
- Single serotype
- Inapparent infection is rare
- No animal reservoir
- Worldwide distribution
- Epidemics in 2 to 3 years
- Leading cause of infant mortality in developing countries
- Incubation 9-11 days
- Infection confers life-long immunity

Pathogenesis

- Humans only known natural host
- Local replication in Upper/lower respiratory tract



- Multinucleated giant cells with intranuclear inclusions in lymphoid tissue
- CNS involvement : Symptomatic encephalitis occurs in 1:1000 cases within 14 to 28 days (autoimmune demyelinating disease)
- Acute progressive infectious encephalitis in patients with defective CMI
- Subacute sclerosing panencephalitis (SSPE) late complication (after 5 to 15 years)
- Atypical measles – previously immunized people with inactivated measles vaccine, exposed to natural infection

Phases of disease:

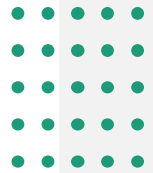
- Incubation period
- Prodromal phase (2-4 days) – fever sneezing, cough, conjunctivitis, Koplik's spots
- Eruptive phase (5 – 7 days) – rash starts on the head, chest, trunk and limbs (discrete maculopapular, coalesce, blotches)
- Atypical measles – high fever, unusual rash, no Koplick's spots
- Modified measles- in infants with residual maternal antibodies, prolonged incubation period, mild prodromal symptoms

Complications:

- Secondary bacterial infections (e.g. β -hemolytic streptococci)
- Otitis media
- Lower Respiratory infection (15%)
- Immunologically deficient children- acute progressive encephalitis ; giant cell pneumonia
- CNS – Electroencephalographic changes in about 50%
- Subacute Sclerosing Panencephalitis (SSPE) – incidence 1:300 000 or 1:1 million; progressive mental deterioration ; fatal

Laboratory diagnosis

- **Specimen**: nasopharyngeal swab, blood
- **inoculation**: Monkey or Human kidney cell lines/ human amniotic cells; Growth 7-10 days
- **Cytopathic effects**: multi nucleated giant cells containing both intranuclear & intraplasmic inclusions
- **Confirmation**: HAd or IF
- **Serology**: Paired sera (four fold rise in Ab titer) demonstration of measles specific IgM, HI, CF, and Nt.

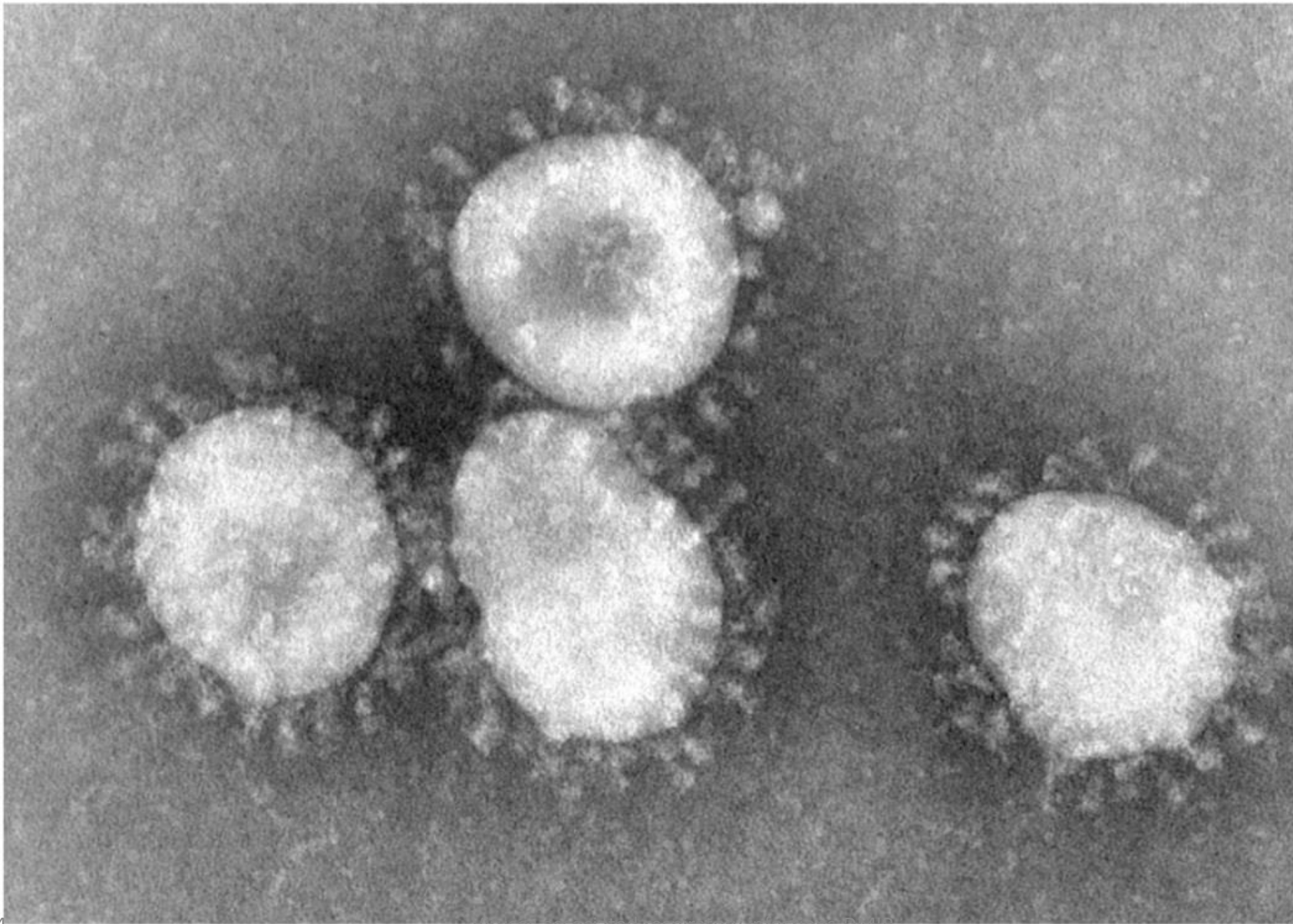


Prevention & Control

- Attenuated live measles virus vaccine
- Highly effective and safe (>95%)
- In 10-15% mild clinical reaction

Coronaviruses

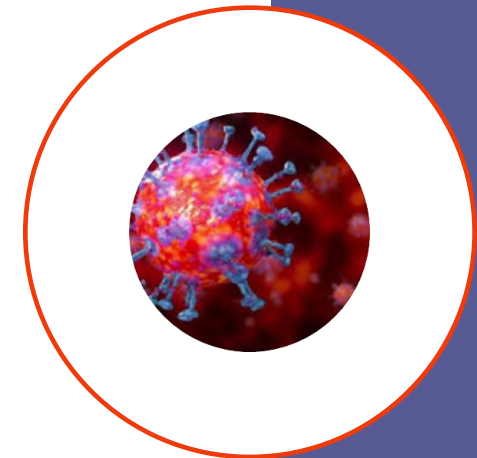
- **Family:** Coronaviridae , Order : Nidovirales
- **Genus :** Beta Coronavirus
- **Genome:** ss RNA, + sense (20-30kb), linear, non-segmented
- **Envelope:** +, bearing club shaped peplomers (spikes), Latin-corona –“crownlike”
- **Shape:** spherical
- **Size:** 80-220nm
- **Nucleocapsid:** Helical
- **Glycoprotein:**
 - **S-** spike protein, receptor binding, cell fusion, major antigen
 - **M-** membrane protein , transmembrane-budding & envelope formation
 - In some types there is a third glycoprotein -> **HE-** hemagglutinin- Esterase;
 - Genome associated with a basic phosphoprotein **N**



- Coronaviruses are classified as a family within the Nidovirales order, viruses that replicate using a nested set of mRNAs ("nido-" for "nest")
- 4 main sub-groupings of coronaviruses :- alpha, beta, gamma, and delta
- The human coronaviruses (HCoVs) are in two of these genera:
 - **alpha coronaviruses** (HCoV-229E and HCoV-NL63) and **beta coronaviruses** (HCoV-HKU1, HCoV-OC43, Middle East respiratory syndrome coronavirus [MERS-CoV], and the severe acute respiratory syndrome coronavirus [SARS-CoV])
- Human coronaviruses were first identified in the mid-1960s

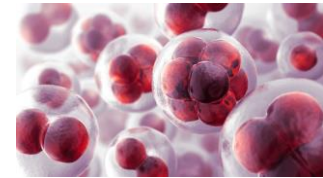
The seven coronaviruses that can infect people are:

- Common human coronaviruses
 - 229E (alpha coronavirus)
 - NL63 (alpha coronavirus)
 - OC43 (beta coronavirus)
 - HKU1 (beta coronavirus)
- Other human coronaviruses
 - **SARS-CoV** (the beta coronavirus that causes severe acute respiratory syndrome, or SARS) – reported in 2003, spread to 26 countries
 - **MERS-CoV** (the beta coronavirus that causes Middle East Respiratory Syndrome, or MERS) –reported 2012, spread to 27 countries
 - **SARS-CoV-2** (the novel coronavirus that causes coronavirus disease 2019, or COVID-19) –reported 2019, an ongoing Pandemic

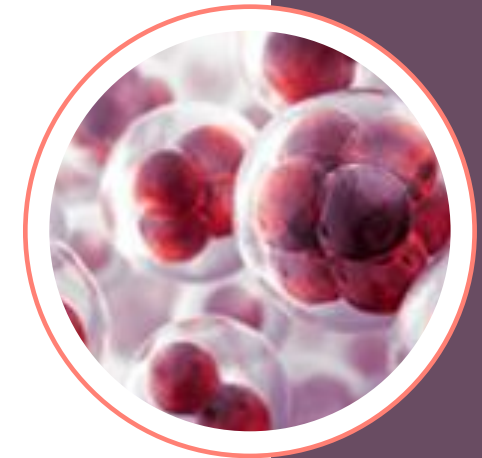


Viral structure and replication

- largest known viral RNA genomes, with a length of 27 to 32 kb.
- The host-derived membrane is studded with glycoprotein spikes and surrounds the genome, which is encased in a nucleocapsid that is helical in its relaxed form but assumes a roughly spherical shape in the virus particle

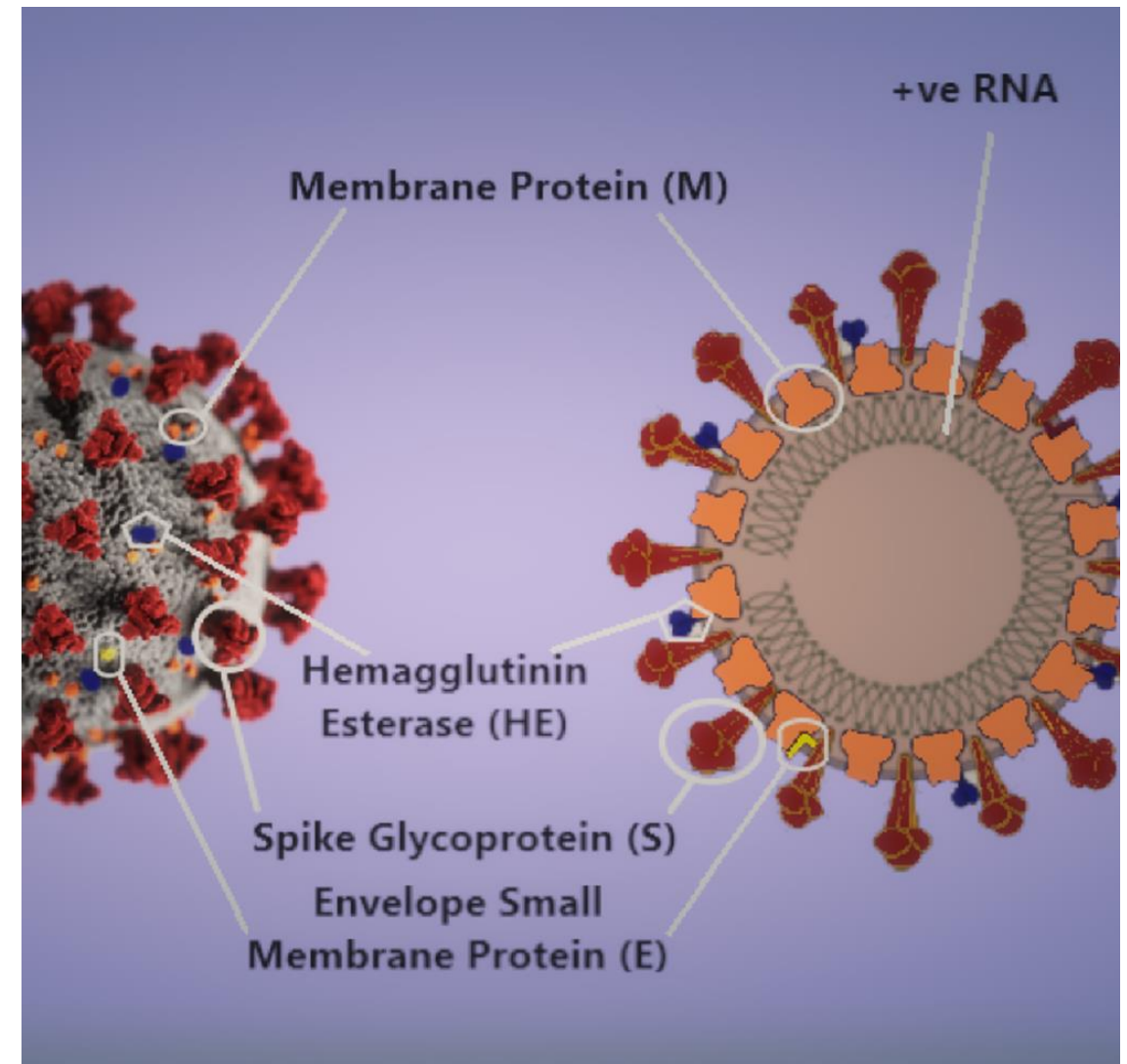


- Replication of viral RNA occurs in the host cytoplasm by a unique mechanism
 - RNA polymerase binds to a leader sequence and then detaches and reattaches at multiple locations
 - allowing for the production of a nested set of mRNA molecules with common 3' ends
- The genome encodes four or five structural proteins, S, M, N, HE, and E



Structural proteins

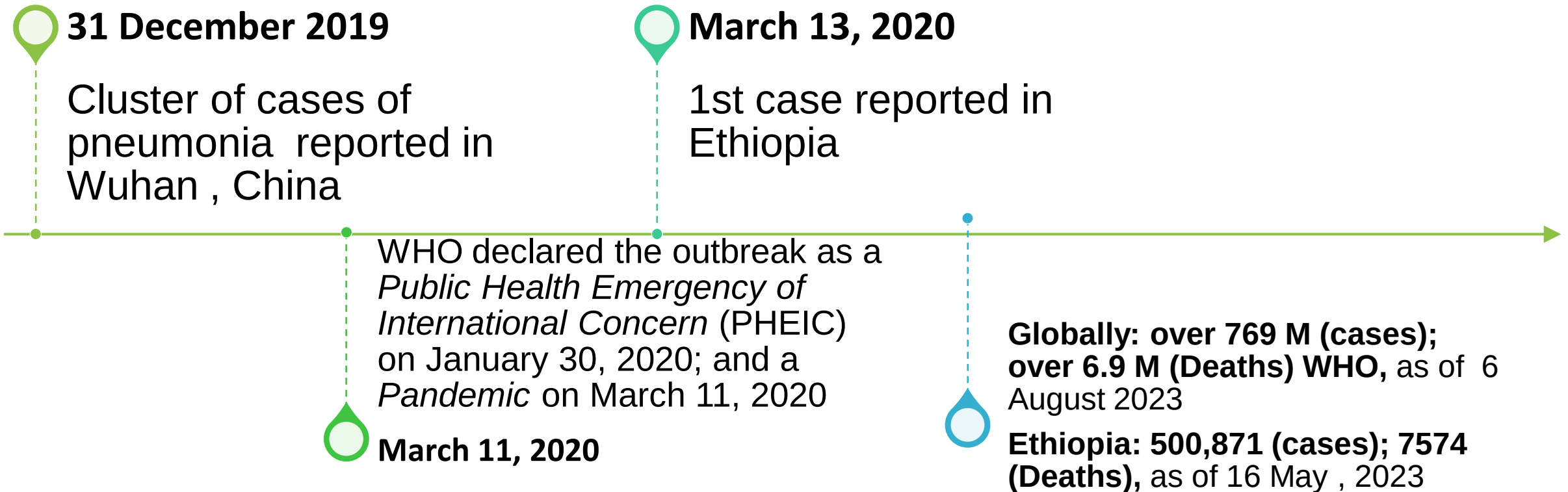
- **N** protein forms a complex with RNA (aids in the regulation of viral synthesis)
- **M** protein projects on the external surface of the envelope (important for viral assembly)
- **E** protein (unclear function)
- **S** protein, club shaped surface projection, gives the virus its characteristic crown like appearance , responsible for receptor binding and fusion with the host cell membrane
 - **S, E and M** proteins create the viral envelope



Ref.: Kakodkar P, Kaka N, Baig M (April 06, 2020) A Comprehensive Literature Review on the Clinical Presentation, and Management of the Pandemic Coronavirus Disease 2019 (COVID-19). Cureus 12(4): e7560. doi:10.7759/cureus.7560

- Coronaviruses are important human and animal viruses
- Research evidence suggests that SARS-CoV and MERS-CoV originated in bats
 - MERS-CoV spreads from infected dromedary camels to people
 - Scientists are trying to determine how SARS-CoV-2 spread from an animal reservoir to people.

Epidemiology of SARS-Cov 2



SARS-CoV-2 variants

From Alpha to.....Omicron

Some of the variants of concern that were being monitored by WHO :

- **Alpha** (B.1.1.7) [first identified in UK , December 2020]
- **Beta** (B.1.351) [first identified in South Africa, December 2020]
- **Gamma** (P.1) [first identified in Brazil, January 2021]
- **Delta** (B.1.617.2) [first detected in India, May 2021]
- **Omicron** (B.1.1.529) (November 2021, detected in many countries rapidly, identified by SA)
- Currently: Three variants of interest (VOIs); **XBB.1.16**, XBB.1.5, and EG.5

Modes of transmission:

- Current evidence suggests that COVID-19 spreads between people through direct, indirect (through contaminated objects or surfaces), or close contact with infected people via mouth and nose secretions. These include saliva, respiratory secretions or secretion droplets (WHO, 2020)

Clinical presentation

- Incubation period 2-14 days
- From asymptomatic to severe pneumonia
- Fever, cough, loss of smell & taste; headache, dyspnea, and others
- ARDS, sepsis
- Complications (acute renal failure; secondary infections)

Laboratory Diagnosis

Safety procedures for appropriate specimen collection, storage, packaging and transport must be ensured

Samples to be collected:

- upper respiratory specimens : nasopharyngeal and oropharyngeal swab or wash in ambulatory patients and/or lower respiratory specimens :
 - sputum (if produced) and/or endotracheal aspirate or bronchoalveolar lavage in patients with more severe respiratory disease

Laboratory testing:

- Nucleic acid amplification tests (NAAT) by real time reverse – transcription Polymerase chain reaction (RT-PCR)
- Serologic testing : detection of IgM or IgG (can aid of an ongoing outbreak and retrospective assessment of the attack rate or extent of an outbreak)
- Rapid tests for antigen detection
- Viral isolation – cell culture
- Sequencing

Other viruses

Upper respiratory infection

Disease	Virus
Rhinitis	Rhinoviruses, adenoviruses, coronaviruses, parainfluenza viruses, influenza viruses, respiratory syncytial virus, some coxsackie A viruses
Pharyngitis or tonsillitis	Adenoviruses, parainfluenza viruses, influenza viruses, rhinoviruses, coxsackie A or B virus, herpes simplex virus, Epstein–Barr virus
Stomatitis	Herpes simplex virus, coxsackie A viruses
4/3/2024	Viruses associated respiratory infections-DMIP



Thank you